

Linear Algebra – Distance, Hyperplanes and Halfspaces, Eigenvalues, Eigenvectors

Hyperplanes

- Geometrically, hyperplane is a geometric entity whose dimension is one less than that of its ambient space.
- For instance, the hyperplanes for a 3D space are 2D planes and hyperplanes for a 2D space are 1D lines and so on.
- The hyperplane is usually described by an equation as follows

$$X^T n + b = 0$$

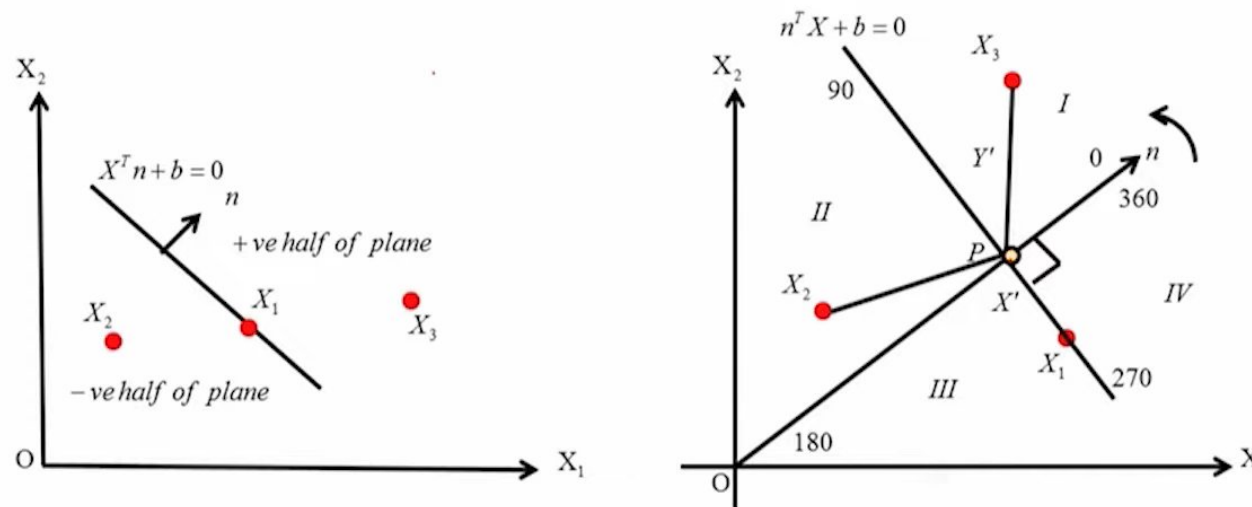
Halfspace

- We can observe that the equation can be evaluated for the two halfspaces
- It can be seen that

$$X^T n + b = 0 \quad \forall X \in \text{line}$$

$$X^T n + b > 0 \quad \forall X \in \text{subspace in the } n \text{ direction } (X_3)$$

$$X^T n + b < 0 \quad \forall X \in \text{subspace in the } -n \text{ direction } (X_2)$$



Halfspace

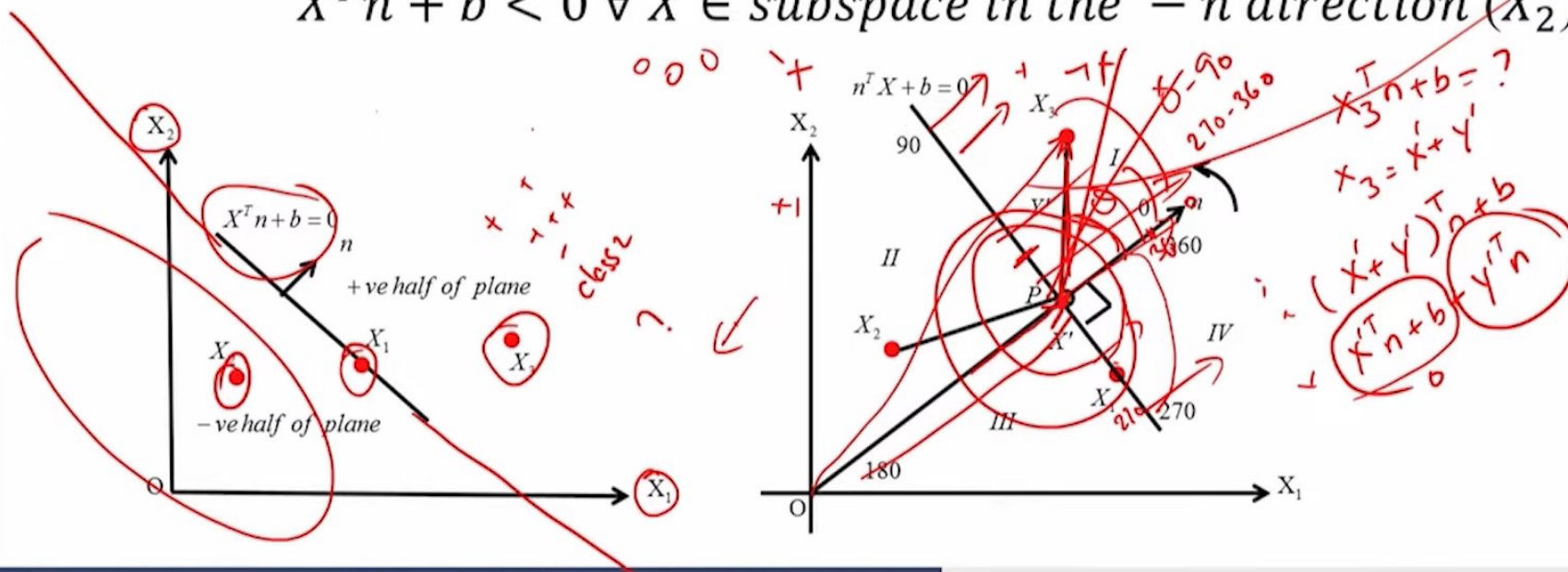
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$$a^T b = |a||b| \cos \theta$$



Hyperplanes and halfspaces: Example

- Let us consider a 2D geometry with $n = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $b = 4$

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$X^T n + b = 0$$

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 4 = 0$$

$$x_1 + 3x_2 + 4 = 0 \quad \checkmark$$

- The hyperplane is the equation of a line
- The halfspaces corresponding to this hyperplane are

$$x_1 + 3x_2 + 4 > 0 : \text{Positive halfspace}$$

$$x_1 + 3x_2 + 4 < 0 : \text{Negative halfspace}$$

Handwritten calculations and points:

- Points: $(-1, -1)$, $(1, -1)$, $(1, -2)$
- Line: $x_1 + 3x_2 + 4 = 0$
- Positive halfspace: $1 - 3 + 4 = 2 > 0$ (for $(1, -1)$)
- Negative halfspace: $1 - 6 + 4 = -1 < 0$ (for $(1, -2)$)

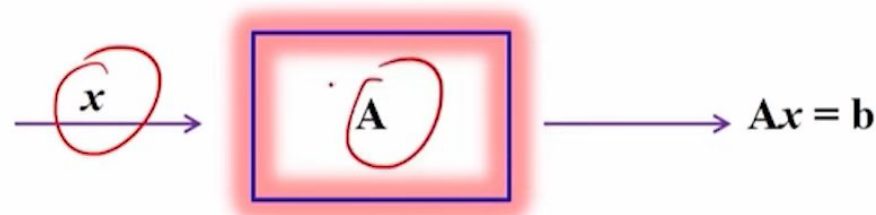
Eigenvalues and eigenvectors

- We have previously seen linear equations of the form $Ax = b$
- What is the geometrical interpretation of this equation?
- We can make an interpretation as follows
 - When vector x is operated on by A , we obtain a new vector b with a different orientation

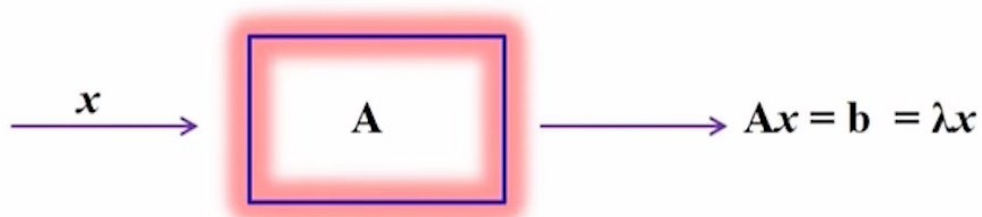


Eigenvalues and eigenvectors

- Operator representation



- The newly obtained b vector represents a new orientation. So we ask the following question
- Are there directions for a matrix A such that when the matrix operates on these directions they maintain their orientation save for multiplication by a scalar (positive or negative)?
- That is

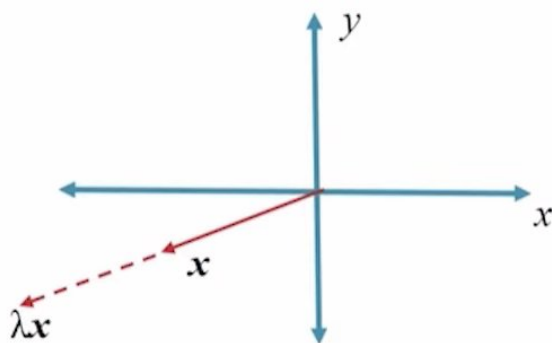


Eigenvalues and eigenvectors

- The mathematical formulation of our question is

$$Ax = \lambda x$$

- The constant λ (*positive*) represents the amount of stretch or shrinkage the attributes x go through in the x direction
- The solutions (x) are known as eigenvectors and their corresponding λ are eigenvalues



Eigenvalues and eigenvectors

- We can find the eigenvalues as follows

$$Ax = \lambda x \quad A(n \times n); x(n \times 1)$$

$$Ax - \lambda Ix = 0$$

$$(A - \lambda I)x = 0$$

- Thus the eigenvalues of the equation can be identified using

$$|A - \lambda I| = 0$$

- Substituting the eigenvalues in the original equation will help us find solutions for the eigenvector x



Eigenvalues and eigenvectors

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$$(A - \lambda I)x = 0 \quad \checkmark$$

$n \times n$ $n \times 1$

$$\begin{pmatrix} \vdots \\ \vdots \\ \vdots \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \end{pmatrix}$$

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$$|A - \lambda I| = 0$$

- Substituting the eigenvalues in the original equation will help us find solutions for the eigenvector x

$x = 0$ (trivial solution)
 x is in the nullspace of $(A - \lambda I)$
 $(\text{rank}) + \text{nullity} = n$

Eigenvalues and eigenvectors: Examples

- Consider the following example with the given A matrix

$$A = \begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \lambda x_1 \\ \lambda x_2 \end{bmatrix}$$

$$|A - \lambda I| = \left| \begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = \left| \begin{bmatrix} 8-\lambda & 7 \\ 2 & 3-\lambda \end{bmatrix} \right|$$

$$= 0$$

$$(8 - \lambda)(3 - \lambda) - 14 = 0$$

$$\lambda^2 - 11\lambda + 10 = 0$$

$$\lambda = (10, 1)$$

- Thus we identify two eigenvalues and now we proceed to find the corresponding eigenvectors

R Code

```
A = matrix(c(8,7,2,3), 2, 2, byrow=TRUE)
ev = eigen(A)
values = ev$values
```

Console output

```
> values
[1] 10 1
```

Eigenvalues and eigenvectors: Examples

$$\begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

- $\lambda = 1$

$$\begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} 8x_1 + 7x_2 \\ 2x_1 + 3x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$x_1 + x_2 = 0$$

- Thus the eigenvector (unit) corresponding to $\lambda = 1$ is

$$X = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$$

Handwritten calculations:

$$8x_1 + 7x_2 = x_1$$

$$7x_1 + 7x_2 = 0 \Rightarrow x_1 + x_2 = 0 \quad \checkmark$$

$$2x_1 + 3x_2 = x_2$$

$$2x_1 + 2x_2 = 0 \Rightarrow x_1 + x_2 = 0$$

$$\sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(-\frac{1}{\sqrt{2}}\right)^2} = \sqrt{\frac{1}{2} + \frac{1}{2}} = \sqrt{1} = 1$$

Eigenvalues and eigenvectors: Examples

- $\lambda = 10$

$$\begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 10x_1 \\ 10x_2 \end{bmatrix}$$

$$\begin{bmatrix} 8x_1 + 7x_2 \\ 2x_1 + 3x_2 \end{bmatrix} = \begin{bmatrix} 10x_1 \\ 10x_2 \end{bmatrix}$$

$$7x_2 = 2x_1 \quad \checkmark$$

- Thus the eigenvector (unit) corresponding to $\lambda = 10$

$$X = \begin{bmatrix} \frac{7}{\sqrt{53}} \\ 2 \\ \frac{2}{\sqrt{53}} \end{bmatrix}$$

K
 $\frac{2K}{7}$

R Code

```
A = matrix(c(8,7,2,3), 2, 2, byrow=TRUE)
ev = eigen(A)
vectors <- ev$vectors
> vectors
```

	[,1]	[,2]
[1,]	0.9615239	-0.7071068
[2,]	0.2747211	0.7071068



Summary

$$Ax = b$$

- Geometric interpretation

$$Ax = \lambda x$$

- Eigenvalue-eigenvector equation

$$\lambda$$

- N eigenvalues from $|A - \lambda I| = 0$

$$x$$

- Eigenvectors, generally expressed in unit vector form

Connections between eigenvectors, column space and null space

- We know that eigenvalues can be complex numbers even for real matrices
- When eigenvalues become complex, eigenvectors also become complex
- However, if the matrix is symmetric, then the eigenvalues are always real
- As a result, eigenvectors of symmetric matrices are also real
- Further, there will always be n linearly independent eigenvectors for symmetric matrices

Handwritten notes:

$$|A - \lambda I| = 0$$

$$P_n(\lambda) = 0$$

$$A\vec{x} = \lambda\vec{x}$$

$$A = A^T$$

$$A = \begin{pmatrix} \lambda_1 & \lambda_2 & \dots & \lambda_n \\ \theta_1 & \theta_2 & \dots & \theta_n \end{pmatrix}$$

$$P_n(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) \dots (\lambda - \lambda_n)$$

$$\lambda = 1 \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Connections between eigenvectors, column space and null space

- Symmetric matrices have a very important role in data sciences
- In fact symmetric matrices of the form $A^T A$ or AA^T are often encountered
- Eigenvalues of matrices of the form $A^T A$ or AA^T while being real are also non-negative
- As discussed for general symmetric matrices, there will be n linearly independent eigenvectors for matrices of this form also
- What is the connection between the eigenvectors and the column space and null space of a (symmetric) matrix ?

$$\begin{aligned} (A^T A)^T &= A^T (A^T)^T \\ &= A^T A \\ &= A^T A \end{aligned}$$

0 →



Connections between eigenvectors, column space and null space

$$Av = \lambda v$$

- What happens when the eigenvalues become zero?

$$Av = 0$$

$AB=0$ ✓
 $\Rightarrow \lambda=0$ is a nullspace vector

- The eigenvectors corresponding to zero eigenvalues are in the null space of the matrix
- Conversely, if the eigenvalue corresponding to an eigenvector is not zero then that eigenvector cannot be in the null space

$Av=0$ and get non-trivial v
 if A is full rank

Connections between eigenvectors, column space and null space

- Let us assume that there are r eigenvectors corresponding to zero eigenvalue
- This means that the null space dimension is r
- From rank-nullity theorem (discussed before), we know that the column rank should be $n - r$
- That is $n - r$ independent vectors are enough to represent all the vectors in the columns of the matrix (column space)
- What could be a basis for this column space or what could be the $n - r$ independent vectors?

*$A_{n \times n}$ is Symmetric
 r zero eigenvalues
 $n - r$ non-zero eigenvalues
 $\rightarrow r$ eigenvectors
 $\text{rank} + \text{nullity} = n$
 $\text{rank} = n - r$*



Connections between eigenvectors, column space and null space

- Notice that there are $n - r$ eigenvectors which are not in the null space
- We know that these are independent
- We also know that these vectors are a linear combination of all the column vectors – that is they are in the column space

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

- Further, we know that the dimension of the column space is $n - r$ (rank-nullity theorem)
- This implies that the eigenvectors corresponding to the non-zero eigenvalues form a basis for the column space

Example

- Consider the following A matrix

$$A = \begin{bmatrix} 0.36 & 0.48 & 0 \\ 0.48 & 0.64 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$A^T = A$$

$$|A - \lambda I| = 0$$

$$P_3(\lambda) = 0$$

$$Ax = \lambda x$$

- Notice that this is a symmetric matrix
- The eigenvalues for this matrix are $\lambda = (0, 1, 2)$
- The eigenvectors corresponding to these eigenvalues are

$$v_1 = \begin{bmatrix} -0.8 \\ 0.6 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0.6 \\ 0.8 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$A v_1 = 0$

3×1
 2×1



Example

- From our prior understanding, the eigenvector corresponding to the zero eigenvalue will be in the null space
- We check that

$$Av_1 = \begin{bmatrix} 0.36 & 0.48 & 0 \\ 0.48 & 0.64 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} -0.8 \\ 0.6 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- Interestingly, in the initial lectures, it was identified that the null space vector identifies a relationship between the variables
- Hence, the eigenvector corresponding to the zero eigenvalue can be used to identify the relationships among variables



Example

- Let us now check if the other two eigenvectors shown below span the column space

$$v_2 = \begin{bmatrix} 0.6 \\ 0.8 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

- This is demonstrated as below

$$\begin{bmatrix} 0.36 \\ 0.48 \\ 0 \end{bmatrix} = 6 * \begin{bmatrix} 0.6 \\ 0.8 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0.48 \\ 0.64 \\ 0 \end{bmatrix} = 8 * \begin{bmatrix} 0.6 \\ 0.8 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} = 2 * \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$



Summary

$$Ax = \lambda x$$

- Symmetric matrices have real eigenvalues

$$Ax = \lambda x$$

- Symmetric matrices also have n linearly independent eigenvectors

$$Ax = 0$$

- Eigenvectors corresponding to zero eigenvalues span the null space

$$Ax = \lambda x$$

- Eigenvectors corresponding to non-zero eigenvalues span the column space for symmetric matrices